OEA r0.2 — 3.3 V / 5 V HOST TO EXTERNAL STEP/DIR/ENABLE

Engineering prototype — not validated for sale or production. Exact host/driver profiles require qualification.

EXT terminal: 5 V / 12 V / 24 V nominal. 5 V terminal input uses EXT only. 5V switch mode requires 12 V or 24 V input.

Switch changes COM+ only; power off before switching. Shared ground, no isolation. VM contact omitted; external power never powers host VIO.

BUCK_VIN

GND

V_EXT_PROT

C1
100nF 50V X7R
V_LOGIC 1 2 GND

C2
100nF 50V X7R
V_LOGIC 1 2 GND

C3
100nF 50V X7R
V_LOGIC 1 2 GND

D1
PMEG6010CEH
VS_SELECT 1 K
A 2 VS_REG

J1
StepStick logic row
EN 1 EN
2 MS1_NC
3 MS2_NC
4 MS3_NC
RST_NC 5
SLP_NC 6
STEP 7 STEP
DIR 8 DIR

J2
StepStick power row; VM physically omitted
GND 1 GND_VM
2 MOTOR_NC1
3 MOTOR_NC2
5 MOTOR_NC3
MOTOR_NC4 5
HOST_VIO 6 HOST_VIO
GND 7 GND

J3
Keyed driver signal output
COM_PLUS 1 COM+
PUL_MINUS 2 PUL−
DIR− 3 DIR_MINUS
ENA− 4 ENA_MINUS

J4
External reference/power 5/12/24V
EXT_RAW 1 EXT+
0V 2 GND

Q1
NX6008NBKW
STEP_GATE 1 G
GND 2 S
D 3 PUL_MINUS

Q2
NX6008NBKW
DIR_GATE 1 G
GND 2 S
D 3 DIR_MINUS

Q3
NX6008NBKW
EN_GATE 1 G
GND 2 S
D 3 ENA_MINUS

R1
220R 1%
STEP 1 2 STEP_IN

R2
220R 1%
DIR 1 2 DIR_IN

R3
220R 1%
EN 1 2 EN_IN

R4
100k 1%
STEP_IN 1 2 GND

R5
100k 1%
DIR_IN 1 2 GND

R6
10k 1%
HOST_VIO 1 2 EN_IN

R7
100k 1%
EN_IN 1 2 GND

R8
220R 1%
STEP_BUF 1 2 STEP_GATE

R9
220R 1%
DIR_BUF 1 2 DIR_GATE

R10
220R 1%
EN_BUF 1 2 EN_GATE

R11
10k 1%
STEP_GATE 1 2 GND

R12
10k 1%
DIR_GATE 1 2 GND

R13
10k 1%
EN_GATE 1 2 GND

R14
100k 1%
LOGIC_READY 1 2 GND

R15
10k 1% 0.125W
COM_PLUS 1 2 GND

SW1
5V / EXT
VS_SELECT 1 5V
COM_PLUS 2 COM
EXT 3 V_EXT_PROT

U1
SN74LVC2G126DCUR
LOGIC_READY 1 1OE
STEP_IN 2 1A
DIR_IN 5 2A
LOGIC_READY 7 2OE
2Y 3 DIR_BUF
1Y 6 STEP_BUF
VCC 8 V_LOGIC
GND 4 GND

U2
SN74LVC1G126DCKR
LOGIC_READY 1 OE
EN_IN 2 A
GND 3 GND
Y 4 EN_BUF
VCC 5 V_LOGIC

U3
TPS3839G33DBZR
GND 1 GND
V_LOGIC 3 VDD
RESET_N 2 LOGIC_READY

C20
2.2uF 100V X7R 10%
BUCK_VIN 1 2 GND

C21
100nF 100V X7R 10%
BUCK_VIN 1 2 GND

C22
1uF 16V X7R 10%
BUCK_VCC 1 2 GND

C23
22uF 25V X7R 20%
VS_REG 1 2 GND

C24
22uF 25V X7R 20%
VS_REG 1 2 GND

C25
220nF 100V X7S 10%
V_EXT_PROT 1 2 GND

C26
4.7uF 25V X7R 10%
V_LOGIC 1 2 GND

D20
28V bidirectional TVS
V_EXT_FUSED 1 1
2 2 GND

D21
12V gate clamp
V_EXT_PROT 1 K
A 2 REV_GATE

D22
60V buck input reverse-current block
BUCK_VIN 1 K
A 2 V_EXT_PROT

F20
0.5A 63V fast fuse
EXT_RAW 1 1
2 2 V_EXT_FUSED

Q20
60V reverse-polarity P-MOSFET
REV_GATE 1 G
V_EXT_PROT 2 S
D 3 V_EXT_FUSED

R20
100k 1%
REV_GATE 1 2 GND

R21
127k 1%
BUCK_VIN 1 2 BUCK_EN

R22
20k 1%
BUCK_EN 1 2 GND

U20
TPSM365R6V5RDNR fixed 5V buck
BUCK_EN 2 EN
BUCK_VIN 3 VIN
5 SW
6 SW
VS_REG 9 BIAS
GND 11 MODE/SYNC
PGOOD 1
VOUT 4 VS_REG
VCC 8 BUCK_VCC
GND 10 GND
BOOT 7

U21
LT3014BES8#TRMPBF -3.34V LDO
V_EXT_PROT 1 IN
GND 2 GND
LDO_ADJ 4 ADJ
OUT 5 V_LOGIC
NC 3

R23
174k 0.1%
V_LOGIC 1 2 LDO_ADJ

R24
100k 0.1%
LDO_ADJ 1 2 GND

R25
22k 1%
V_LOGIC 1 2 GND

Signal connector: 1 COM+ | 2 PUL− | 3 DIR− | 4 ENA−. Common-positive drivers only; configure voltage/current and enable semantics per exact driver manual.

Native ERC checks electrical declarations; it does not establish loaded timing, startup behavior, protection performance, host fit or physical qualification.

External signal supply with 5V / EXT selector
ENGINEERING — NOT RELEASED
Sheet: /
File: adapter.kicad_sch
Title: OEA external-driver adapter
Size: A2 | Date: 2026-10-06 | **Rev: r0.2**
KiCad E.D.A. 10.0.4 | Id: 1/1